Aswith Reddy Sama

(773) 673-4072 | aswithreddy907@gmail.com | LinkedIn | GitHub | Portfolio | USA

PROFESSIONAL SUMMARY

AI Engineer with **3+ years** of combined **AI/ML** and **software engineering experience**, specializing in production-grade LLM applications, multi-agent systems, and RAG. Experienced in building end-to-end AI applications with deterministic workflows and guardrails, fine-tuning transformer models with PEFT/LoRA, and developing scalable backend services with Python, FastAPI, PostgreSQL, Docker, AWS, and GCP.

PROFESSIONAL EXPERIENCE

Software Engineering (AI/ML) Intern | Box Technology Group **August 2026 – Present**

*Project: **Ressa – Multi-Agent Conversational AI Platform** | OpenAI, RAG, FastAPI, LangChain* Delaware, USA

- **Owned multi-agent AI architecture**, orchestrating RAG, messaging, CRM and scheduling across **16 real estate domains**.
- **Built** an **OpenAI-powered reasoning engine** for **intent classification**, **fact extraction**, and **context-aware decisions** across **7 action classes**.
- Developed **tenant-safe FastAPI services**, integrating **Twilio**, **Gmail**, & **Microsoft Graph** for **messaging**, **calls**, and **scheduling** across **3 channels.**
- Designed **LLM evaluation framework** for **320+** scenarios, targeting **95%** classification accuracy & **90%** context retention with **AI guardrails**.

AI Engineer | Catapult **May 2026 – July 2026**

*Project: **Intelligent-Multi-Tenant-Messaging-System** | LangGraph, OpenAI, FastAPI* Utah, USA

- Led and Architected a 3-layer **LangGraph-based multi-agent AI** system automating customer service, classifying inbound messages across Upsell, Billing, After-Service, and Review with **15+ sub-domain categories** guiding **LLM decision-making** for 70+ pest-control companies.
- Engineered a **deterministic AI framework** with production **guardrails,** restricting LLMs to approved workflows while separate execution logic handled **tool calling** and PostgreSQL updates, reducing hallucinations and ensuring safe, **consistent** automation.
- Engineered an **AI pipeline** using **GPT-5-mini hybrid reasoning**, FastAPI, Twilio, and CRMs, with **AI-worker microservices** on **GCP Cloud Run.**
- Collaborated with CEO on **prompt & context engineering**, **optimizing AI agent's decision context**, achieving **~95% production accuracy**.

Data Scientist | Chicago Education Advocacy Cooperative (NGO) **February 2026 – May 2026**

***Project:** Talent Recruitment Platform | OpenAI, Next.js, Supabase, PostgreSQL* Chicago, USA

- Built an **AI-powered recruitment platform** using **GPT-5-mini**, automating resume parsing, and extraction of candidate skills, experience.
- Engineered a **full-stack candidate discovery system with 12**+ search filters, **role-based access controls**, and **PostgreSQL**, integrating automated deployment through **AWS ECS** and **GitHub Actions**.

AI/ML Algorithm Engineering Intern | Uplifty **July 2025 – November 2025**

*Project: **Human Wellness NLP Pipeline** | LLM Orchestration, PyTorch, BERTweet* New York, USA

- **Spearheaded** the **Human Growth Index**, designed a multi-dimensional well-being measure and a **multi-agent LLM weak-labeling pipeline** annotating **100K+ texts**, routing ~8% conflicting GPT-4o-mini classifications to GPT-4.1, escalating unresolved cases to human review.
- **Fine-tuned** BERTweet with **PEFT/LoRA** on the curated dataset, achieving **0.70–0.90 F1 scores** across 27 emotion and 13 well-being categories; transformed predictions into explainable well-being score and deployed production APIs using Flask & GCP.

Software Engineer / Analyst | Unizen Technologies Pvt. Ltd. **May 2022 – December 2023**

*Project: **Learning Management System (LMS) Backend Platform** | Node.js, Express.js, AWS* Bangalore, India

- Developed **20+ REST APIs** using **Node.js, Express.js, & MySQL** for LMS workflows, supporting **5000+ users** & reducing API latency by **~20%**.
- Integrated **Amazon S3** for secure document storage and automated **CI/CD pipelines** using **AWS CodePipeline**.

*Project: **Enterprise Data Pipeline** | PySpark, PowerBI, scikit-learn*

- Transformed **100K+ sales and inventory records** using **Databricks (PySpark)** for analytics, demand forecasting, and **Power BI** reporting.
- Assisted in building **Azure Data Factory ETL pipelines**, integrating **SAP ERP, SQL Server, and flat-file data** into the analytics platform.

PROJECTS

***Computer Use Automation** | Interface.ai | Python, Playwright, Pydantic* **September 2026 – October 2026**

- Architected an **AI-powered browser automation system** using **Playwright**, enabling autonomous discovery, **5 reusable capabilities**, & **LLM-free deterministic replay** with **AI guardrails**, policy enforcement, & human assisted recovery, validated by **23 real-browser acceptance tests**.
- Engineered GPT-5-mini-powered workflow discovery with **SHA-256 browser-state fingerprinting**, **trajectory optimization**, and **parameterized input/output bindings**, converting exploratory LLM actions into verified, reusable execution workflows.

***RAG-Powered Policy Retrieval System** | Hack Midwest | Python, MongoDB Atlas, BGE Embeddings* **October 2025 – November 2025**

- Built **multi-stage RAG pipeline** for health-insurance Q&A, processing PDFs with **PyMuPDF, Docling** into **38 chunks** & **105 sections.**
- Engineered **3-level hybrid hierarchical retrieval** using **768-D BGE embeddings, MongoDB Atlas Vector Search, BM25, RRF, and MiniLM CrossEncoder reranking**, achieving **~85% Recall@3 across 50+ queries** with grounded **GPT-5-mini** responses.

SKILLS & CERTIFICATIONS

- **Languages:** Python, Java, C++, SQL.
- **AI Engineering:** LLMs, OpenAI API, Multi-Agent Architectures & Orchestration, ReAct, LangChain, LangGraph, Prompt & Context Engineering, Deterministic AI Workflows, Tool Calling, AI Guardrails, Vector DBs, RAG, Hybrid Retrieval, Cross-Encoder Reranking, MCP.
- **Deep Learning:** TensorFlow, PyTorch, NLP, Transformer Models, PEFT/LoRA, CNNs, RNNs, YOLOv3.
- **Backend & Cloud:** FastAPI, REST APIs, PostgreSQL, Node.js, Kubernetes, Docker, Git, CI/CD Pipelines, AWS (S3, Bedrock), GCP (Cloud Run).
- **Certifications:** AWS Certified Machine Learning Associate, Mathematics for Machine Learning.

EDUCATION

Illinois Institute of Technology | Chicago, USA **January 2024 – December 2025**

Master of Science in Artificial Intelligence — GPA: 3.70/4.00

J.B. Institute of Engineering and Technology | Hyderabad, India **July 2019 – May 2023**

Bachelor of Technology in Information Technology — GPA: 3.57/4.00